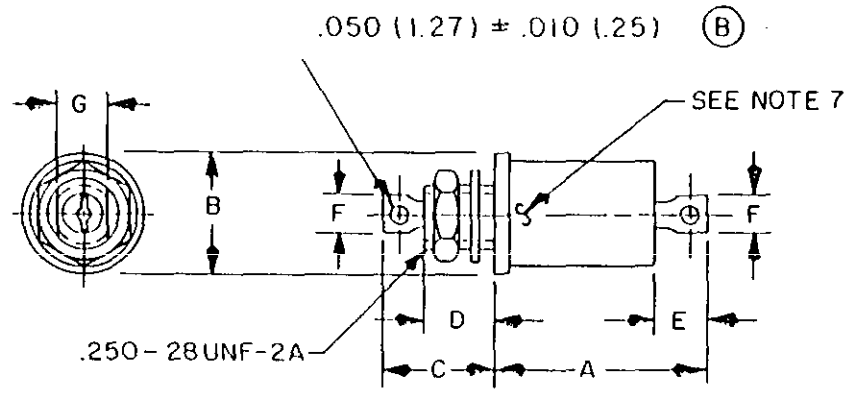
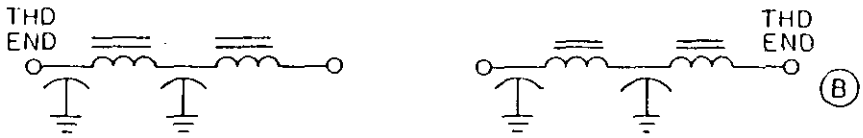
MILITARY SPECIFICATION SHEET

FILTER, RADIO FREQUENCY INTERFERENCE, HERMETICALLY SEALED STYLE FL61

This specification is approved for use by Aeronautical Systems Division, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the filters described herein shall consist of this document and the latest issue of Specification MIL-F-15733.





CIRCUIT CONFIGURATION (2L)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
4. Circuit diagram is for information only.
5. All filters shall be supplied with mounting hardware consisting of a phosper-bronze internal tooth lockwasher, and a brass hexagonal nut, both shall be silver plated.
6. Use of style FL61 with or without shoulder is optional.
7. Terminal identification (nonsymmetrical filters): The case shall be marked at the threaded end of the filter, with the symbol, "L".

FIGURE 1. Case and circuit configuration.

(B) Denotes changes

FSC 5915

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TABLE I. Dimensions.

Type designation 1/ 2/ 3/	Dash number	A	B		C	D		E	F		G	
		Max	Min	Max	Max	Min	Max	Max	Min	Max	Min	Max
FL61YW1ED3	0001	1.050 (26.67)	.390 (9.91)	.410 (10.41)	.352 (8.94)	.177 (4.50)	.197 (5.00)	.190 (4.83)	.085 (2.16)	.105 (2.67)	.196 (4.98)	.206 (5.23)
FL61BW1ED3	0002	1.050 (26.67)	.390 (9.91)	.410 (10.41)	.352 (8.94)	.177 (4.50)	.197 (5.00)	.190 (4.83)	.085 (2.16)	.105 (2.67)	.196 (4.98)	.206 (5.23)
FL61WW1ED3	0003	1.09 (27.7)	---	.416 (10.57)	---	.180 (4.57)	.200 (5.08)	---	.085 (2.16) REF	.105 (2.67) REF	.196 (4.98)	.206 (5.23)

1/ The type designation does not describe a discrete item. For complete identification, the part number and type designation should be referenced.
2/ The letter "W" indicates insertion loss versus frequency characteristics specified in table II.
3/ The letter "D" indicates the operating temperature range -55° to +125°C.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Case: Metal.

Weight: 8.1 grams, maximum (0001 and 0002).
6.795 grams, maximum (0003).

Terminals: Solderable.

Temperature characteristic: D - - - - - -55° to +125°C.

Operating temperature range: -55° to +125°C.

Rated voltage: See table II.

Rated current: See table II.

Insertion loss characteristic: Symbol W (see table II).

Creepage and clearance: Not applicable.

Seal: Method 112, test condition A, MIL-STD-202.

Capacitance to ground: Not applicable.

Temperature rise: 25°C, maximum.

Dielectric withstanding voltage: Method 301, MIL-STD-202.
Test temperature: 25°C.
Test voltage: 100 Vdc applied between each terminal and ground, for a period of 1 to 5 seconds.
Maximum charging current: 50 mA.

Barometric pressure (reduced):
Method 105, test condition B (50,000 ft), MIL-STD-202.
Dielectric withstanding voltage: 125 percent of the rated voltage.
Maximum charging current: 50 mA.

Insulation resistance: Method 302, test condition A, MIL-STD-202.
Test temperature: 25°C.
Test potential: 200 percent of rated dc voltage.
Maximum charging current: 50 mA.
Insulation resistance shall be not less than 1,000 megohms.

Voltage drop: See table II.

Insertion loss: In accordance with MIL-STD-220 and table II.
Insertion loss (check test): In accordance with MIL-F-15733.

Overload: Filters shall be suspended by their terminals in free air at an ambient temperature of 25° ±5°C. A test current of 140 percent of rated current shall be applied for 15 minutes. After the filter has returned to room temperature, insulation resistance and voltage drop measurements shall meet initial requirements.

Terminal strength: Method 211, test condition A (pull), MIL-STD-202.
Force: 5 pounds.

Resistance to soldering heat: Method 210, test condition B, MIL-STD-202.

Salt spray (corrosion): Method 101, test condition A, MIL-STD-202.

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TABLE II. Electrical characteristics.

Dash number	Circuit configuration	DC resistance (ohms)	Maximum rated current (amps)	Rated dc voltage (volts)	Maximum voltage drop (volts)	Minimum full-load insertion loss (dB) in accordance with MIL-STD-220 at 25°C
						Above 1 MHz
0001	$2L_1$	--	2.0	50	--	80
0002	$2L_1$	1.5	.3	50	.45	80
0003	$2L_2$	--	.1	50	.5	55 dB at 50 kHz, 90 dB from 120 kHz to 1 GHz

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Thermal shock and immersion:
Thermal shock: Method 107, test condition A, MIL-STD-202, except the temperature at step 3 shall be +125°C.
Immersion: Method 104, test condition A, MIL-STD-202.
Measurements after final cycling:
Dielectric withstanding voltage with 90 percent of the initial voltage applied for 5 ±1 seconds.
Maximum charging current: 50 mA.
Insulation resistance shall be not less than 30 percent of the initial requirement.
Insertion loss (check test) shall be measured and shall meet the initial requirements.

Shock (specified pulse): Method 213, test condition I, MIL-STD-202.
Electrical load: 100 percent of the rated voltage between terminals and case.

Vibration, high frequency: Method 204, test condition B (20 G), MIL-STD-202.
Electrical load: Rated voltage and current as specified in table II, shall be applied continuously during test.
Test during vibration: During the last cycle in each direction, an electrical measurement shall be made to determine intermittent open or short-circuiting.
After the vibration test, filters shall be subjected to and shall meet the initial requirements of the insertion loss (check test).

Moisture resistance: Method 102, test condition D, MIL-STD-202, except that the temperature at step 3 shall be +125°C. No measurement shall be made before and after cycling. Filters shall then be tested in accordance with method 106, MIL-STD-202.
Initial measurements: Not applicable.
Polarization voltage: 28 Vdc.
Measurements after 24-hour drying period:
Dielectric withstanding voltage with 90 percent of the initial voltage applied for 5 ±1 seconds. Maximum charging current: 50 mA.
Insulation resistance shall be not less than 50 megohms.
Insertion loss (check test) shall be measured and shall meet the initial requirements.

Life: Method 108, MIL-STD-202. Test condition D (1,000 hours) for qualification inspection. Test condition B (250 hours) for group C inspection. Test temperature +125°C.
Test voltage: 1.4 times rated dc voltage.
Measurements after test:
Dielectric withstanding voltage with 90 percent of the initial voltage applied for 5 ±1 seconds. Maximum charging current: 50 mA.
Insulation resistance shall be not less than 30 percent of the initial requirement.
Insertion loss (check test) shall be measured and shall meet the initial requirements.

Part number: M15733/56- (dash number from table I).

Supersession data: This specification includes the requirements of contractor (00724), specification 08-00804 revision D, dated 14 November 1973, contractor (05869) Drawing 2938671, dated 5 May 1972.

Preparing activity:
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